Endalkachew Abebe Kebede
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University of Delaware
205 Pearson Hall | Newark, DE 19716
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EDUCATION

August 2026 **PhD in Geography**
University of Delaware, Newark, Delaware
Dissertation: Global Irrigation, Water Scarcity, and Food Sustainability in the 21st Century

July 2022 **Erasmus Mundus Joint Master's Degree in Soil Science**
The University of Agriculture in Krakow, Poland, the Agricultural University of Plovdiv, Bulgaria, and Ondokuz Mayis University, Samsun, Turkey
Thesis: Soil Fertility Assessment Using Remote Sensing and In-Situ Physiological Analyses of Wheat.

April 2021 **Master of Science in Water Science and Engineering, specializing in Hydraulic Engineering and River Basin Development**
IHE Delft Institute of Water Education, Delft, The Netherlands
Thesis: 1D morphological adaptation of the Lower Zambezi River to dam construction

July 2015 **Master of Science in Irrigation Engineering**
Haramaya University, Haramaya, Ethiopia
Thesis: A Comparison of Hydrological Models Under Climate Change on the Water Resources of Megech River Catchment, Abbay Basin, Ethiopia

July 2009 **Bachelor of Science in Soil and Water Engineering and Management**
Haramaya University, Haramaya, Ethiopia

RESEARCH INTERESTS

Global Environmental Change; Remote Sensing and Earth Observation; Geospatial Data Science; Hydrology and Water Resources; Irrigation and Agricultural Water Management; Food–Water Systems.

RESEARCH AND TEACHING EXPERIENCE

August 2026–Present **Postdoctoral Researcher**, Department of Geography and Spatial Sciences, University of Delaware, USA

- Lead independent research on satellite-based monitoring of irrigated agriculture, extending GMIA-NEXT from a static global map to a high-resolution time series of irrigation extent
- Develop machine learning and data-fusion workflows combining Landsat and Sentinel imagery, agricultural statistics, and cloud and high-performance computing to track irrigation expansion, contraction, and seasonal timing
- Lead manuscripts and open-source data releases, and contribute to research proposals and collaborations with national and international partners
- Mentor graduate and undergraduate researchers in remote sensing, machine learning, and reproducible geospatial workflows

2022- 2026 **Research Assistant,** Department of Geography and Spatial Science, University of Delaware, USA

- Led the development and publication of MIRCA-OS, a global open-source dataset of monthly irrigated and rainfed cropped areas
- Led the development of GMIA-NEXT, a high-resolution (30 m) global irrigated area dataset using integrated remote sensing, machine learning, and geospatial analysis
- Led the update and expansion of the global Area Equipped for Irrigation (AEI) dataset (2000–2020), providing an updated spatially explicit global inventory of irrigated areas for agricultural and water resources research
- Conducted global and local assessments of water scarcity impacts on food production and supply chains, and investigated the socioeconomic and environmental drivers of agricultural economic water scarcity in developing regions
- Evaluated and optimized rental payment structures in the Conservation Reserve Program (CRP) to maximize environmental benefits
- Co-authored interdisciplinary research on agricultural and water sustainability, including the development of the HarvestStat dataset

2017-2022 **Researcher and Lecturer**, Guna Tana Integrated Field Research and Development Center, Debre Tabor University, Ethiopia. (75% research, 25% teaching)

- Designed and led problem-driven research projects in agricultural water management and water resources modeling
- Served as Principal Investigator for two funded projects, and worked as co-PI for multiple eco-hydrology projects
- Led large-scale community service initiatives addressing critical, prioritized local development needs
- Taught undergraduate courses, including Soil and Water Conservation and Introduction to Natural Resource Management

2012-2017 **Lecturer**, Department of Soil and Water Resource Management, Woldia University, Ethiopia. (75% teaching, 25% research)

- Served as Interim Department Head for one year, overseeing all academic and administrative operations, and as a member of the Academic Council
- Taught undergraduate courses including Surface and Ground Water Hydrology, Soil Physics, Irrigation Engineering, GIS, and Remote Sensing
- Supervised undergraduate research projects, coordinated seminars, and advised students on senior theses
- Contributed to departmental curriculum planning

OTHER APPOINTMENT

2010-2012 **Instructor,** Bahir Dar Polytechnic College, Department of Water Technology, Ethiopia.

2010 - 2010 **Irrigation Infrastructure Engineer**, Amhara State Bureau of Agriculture Irrigation Directorate, Ethiopia.

PUBLICATIONS

Published Manuscripts

2026 **Kebede, E.A.,** Hrozencik, A., Marston, L., Van Meter, K., Wei, D., Davis, K.F (2026). Optimizing cropland conservation payments to minimize nitrogen surplus in the Mississippi River Basin. Environ. Res. Lett. 21 104006. https://doi.org/10.1088/1748-9326/ae6671

Gum, J., Caprarelli, G., Montanari, A., Sedora, B., Stall, S., … **Kebede, E.A.,** … Davis, K.F., … Minisini, D. (2026). Impactful datasets in Earth, Space and Environmental Sciences: Highlighting the data that transform research. AGU Advances, 7. https://doi.org/10.1029/2026AV002623

2025 **Kebede, E.A.,** Oluoch, K.O., Siebert, S., Mehta, P., Hartman, S., Jägermeyr, J., Ray, D., Ali, T., Brauman, K.A., Deng, Q., Xie, W., Davis, K.F. (2025). A global open-source dataset of monthly irrigated and rainfed cropped areas (MIRCA-OS) for the 21st Century. *Sci Data* 12, 208. https://doi.org/10.1038/s41597-024-04313-w

Lee, D., Anderson, W., Chen, X., **Kebede, E.A**., Flach, R., Meyer, C., Proctor, J., Ray, D., You, L., Foley, M., Kerdiles, H., Hultgren, A., Huybers, P. (2025). HarvestStat Africa – Harmonized Subnational Crop Statistics for Sub-Saharan Africa. *Sci Data* 12, 690. https://doi.org/10.1038/s41597-025-05001-z

Davis K.F., Anderson, W., Ehrmann, S., Flach, R., Meyer, C., Proctor, J., Ray, D., You L., Foley, M., Kerdiles, H., Hultgren, A., Huybers, P., **Kebede, E.A.**, Meroni M., Park, C., Lee, D. (2025). HarvestStat: A global effort towards open and standardized sub-national agricultural data. *Environmental Research Letters*. http://iopscience.iop.org/article/10.1088/1748-9326/adcb54

Wang, L., Garland, G.M., Ge, T., Guo, S., **Kebede, E.A.,** He, C., Hijri, M., Plaza-Bonilla, D., Stringer, L.C., Davis, K.F., Lee, S.-J., Feng, S., Wang, L., Wei, Z., Cao, H., Wang, Z., Xu, J., Siddique, K.H., Gan, G.Y., Zhao, M. (2025). Integrated strategies for enhancing agrifood productivity, lowering greenhouse gas emissions, and improving soil health. *The Innovation.* https://doi.org/10.1016/j.xinn.2025.101006 .

2024 **Kebede, E.A.,** Abou Ali, H., Clavelle, T., Froehlich, H.E., Gephart, J.A., Hartman, S., Herrero, M., Kerner, H., Mehta, P., Nakalembe, C., Ray, D.K., Siebert, S., Thornton, P., Davis, K.F. (2024). Assessing and addressing the global state of food production data scarcity. *Nat Rev Earth Environ* 5, 295–311. https://doi.org/10.1038/s43017-024-00516-2

Davis, K.F., Abou Ali, H., **Kebede, E.A.**, Khan, B., Sarwar, A. (2024). Where global crop yields may falter next. *Nat Food* 5, 98–99. https://doi.org/10.1038/s43016-023-00911-w

Kebede, E. A., Vasileva, S., Ivanov, B., Dengiz, O., Bojinov, B. (2024). Optimizing data collection in precision agriculture – comparing remote sensing and in situ analyses. *Bulgarian Journal of Agricultural Science*, 30(1), 11–16.

2023 Hagos, Y.G., Andualem, T.G., Sebhat, M.Y., **Kebede, E.A.** (2023). Soil erosion estimation and erosion risk area prioritization using GIS-based RUSLE model and identification of conservation strategies in Jejebe watershed, Southwestern Ethiopia. *Environ Monit Assess* 195, 1501. https://doi.org/10.1007/s10661-023-12136-2

2022 Hagos Y.G., Andualem, T.G., Yibeltal, M., Malede, D.A., Melesse, A.M., Teshome, F.T., Bayabil, H.K., **Kebede, E.A.,** Demissie, E.A., Mitku, A.B. (2022). Assessment of Agricultural Land Suitability for Surface Irrigation Using Geospatial Techniques in the Lower Omo-Gibe Basin, Ethiopia. *Water* 14, 3887. https://doi.org/10.3390/w14233887

2017 **Kebede EA.,** Kebede A (2017). Assessment of Climate Change Impacts on the Water Resources of Megech River Catchment, Abbay Basin, Ethiopia. Open Journal of Modern Hydrology 7, 141-152. https://doi.org/10.4236/ojmh.2017.72008

Manuscripts Under Review and in Preparation

2026 **Kebede, E.A.,** Zhu, A., Citrini, A., Demeke, B., Mialyk, O., Mekonnen, M.M., Chiarelli, D.D., Rulli, M.C., Rosa, L., Su, H., Ali, T., Xie, W., Jägermeyr, J., Brauman, K.A., Deng, Q., Fu, Z., Hartman, S., Kummu, M., Marston, L.T., Mehta, P., Ray, D.K., Richter, B.D., Siebert, S., Tuninetti, M., Liu, W., Davis, K.F. (2026). Exposure of Global Nutrition to Rising Water Scarcity. (Under review at *Science*).

Kebede, E. A., Xie, Y., Laboy, G., Bhimani, K., Boser, A., Urfels, A., Khan, B., Adepoju, M., Adeluyi, O., Brauman, K. A., Casirati, S., Chandrasekaran, S., Deines, J. M., Flach, R., Hansen, M. C., Hartman, S., Jobbagy, E., Khan, A., Kurapati, V., Lark, T., Marquez, J., Michael, H., Nakalembe, C., Ng, K. H., Ng, K. W., Nozari, S., Potapov, P., Rosa, L., Siebert, S., Smith, R., Taufer, M., & Davis, K. F. (2026). GMIA-NEXT: Next-Generation Global Map of Irrigated Areas Using Data Fusion. (Under review at *Scientific Data*). https://doi.org/10.21203/rs.3.rs-10085674/v2

Marston, L., Lamsal, G., **Kebede, E.A.**, Khan, B., Uponi, J., Davis, K.F (2026). Reinvesting water savings to increase U.S. food production. (Under review at *Nature Water*). https://doi.org/10.21203/rs.3.rs-2160460/v1

Mehta, P., Müller, M., **Kebede, E.A.**, Khan, B., Uponi, J., Boser, A., Schmitt, R., Davis, K.F (2026). Have irrigated areas actually expanded around new irrigation dams?. (Under review at *Communications Earth and Environment*).

Davis, K.F.**, Kebede, E**.**A**., Citrini, A., Demeke, B., Mialyk, O., Mekonnen, M.M., Rosa, L (2026). The shifting geography and drivers of global blue water gaps. (*In Prep*).

Kebede, E.A., Davis, K.F., Citrini, A., Rosa, L (2026). Evolution and drivers of global agricultural economic water scarcity. (*In Prep*).

INVITED TALKS

2026 **Kebede, E.A.** (2026). From Fields to Plates: Global Agricultural Water Scarcity and Dietary Nutrient Supply. Fall 2026 Speaker Series, Department of Geography and Spatial Sciences, University of Delaware. In-person (Invited).

Kebede, E.A. (2026). Advancing Global Irrigation Mapping: From Foundational Datasets to High-Resolution Time-Series Monitoring. Global Water Security Center, the University of Alabama. Virtual (Invited).

2025 **Kebede, E.A.** (2025). Advancing High-Resolution Irrigation Mapping with Open-Source Data and Machine Learning, Global Computing Laboratory, University of Tennessee. Virtual (Invited).

Davis, K. F., & **Kebede, E.A.** (2025). The role of irrigation for sustainable and resilient food systems, Oral Presentation. Global Water Security Center. University of Alabama. Virtual (Invited).

Davis, K. F., & **Kebede, E.A.** (2025). The role of irrigation for sustainable and resilient food systems, Oral Presentation. Land and Carbon Lab. World Resources Institute. Virtual (Invited).

2024 Davis KF, **Kebede, E.A.,** Khan B, Mehta P, Sarwar A. (2024) Towards sustainable and resilient food-water systems, Oral Presentation. AGU Global Environmental Change Early Career Award Webinar Series. Virtual (Invited).

DATASETS

2026 **Kebede, E.A.,** Xie, Y., Laboy, G., Bhimani, K., Boser, A., Urfels, A., Khan, B., Adepoju, M., Adeluyi, O., Brauman, K. A., Casirati, S., Chandrasekaran, S., Deines, J. M., Flach, R., Hansen, M.C., Hartman, S., Jobbagy, E., Khan, A., Kurapati, V., Lark, T., Marquez, J., Michael, H., Nakalembe, C., Ng, K.H., Ng, K.W., Nozari, S., Potapov, P., Rosa, L., Siebert, S., Smith, R., Taufer, M., and Davis, K.F. (2026). GMIA-NEXT: Next-Generation Global Map of Irrigated Areas. Zenodo

2026 **Kebede, E.A**., Mehta, P., Siebert, S., Davis, K.F. (2026). Have irrigated areas actually expanded around new irrigation dams? Zenodo

2025 **Kebede, E.A**, Mehta, P., Siebert, S., Davis, K.F. (2025). Global Area Equipped for Irrigation (AEI) for the 21st Century. Zenodo.

2024 **Kebede, E.A.**, Oluoch, K., Siebert, S., P. Mehta, Hartman, S., J. Jägermeyr, D. Ray, T. Ali, K. A. Brauman, Q. Deng, W. Xie, K. F. Davis (2024). A global open-source dataset of monthly irrigated and rainfed cropped areas (MIRCA-OS) for the 21st Century. *HydroShare.*

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FUNDED PROJECTS

Grant Proposals Submitted (Funded)

2023 **Kebede, E.A (**Principal Investigator). A Global Dataset of Monthly Irrigated and Rainfed Cropped Areas (MIRCA) for the 21st Century. CUAHSI Hydroinformatics Innovation Fellowship (Funded). **$5,000**.

2019 Wassi, H.M., Addise, M.B., **Kebede, E.A.**, Gelaye, T.M., Mebrat, W., Alemu, A. (Co-Principal Investigator). Participatory Eco-Hydrology Project on the Gumara Catchment, Upper Blue Nile Basin, Ethiopia. Ethiopian Ministry of Water Resources. **1 Million ETB /40,000 USD.**

2018 **Kebede, E.A.,** Addisse, MB, Abiyu, A. (Principal Investigator). Improving Agricultural Water Productivity in the Guna Tana Watershed. Debre Tabor University, Guna Tana Integrated Field Research and Development Center. **500,000 ETB /18518 USD.**

2017 **Kebede, E.A.** (Principal Investigator), Gebrie, T., Getachew, G., Addise, MB. Hydrodynamic Modeling of the Upper Ribb Catchment for Increased Sustainability of Reservoirs through Community-Managed Catchments. Debre Tabor University, Guna Tana Integrated Field Research and Development Center. **300,000 ETB /11100 USD.**

Grant Proposals Submitted (Not Funded)

2025 Davis, K.F. (PI), **Kebede, E.A.** (Co-PI). AI-Enabled Mapping of Global Irrigated Areas. Delaware NASA EPSCoR Research Infrastructure Development (RID) Seed Grant Program.

Davis, K.F. (PI), **Kebede, E.A.** (Co-PI). Identifying Economically Feasible and Environmentally Sustainable Locations for Switchgrass Production in the U.S. Northeast. Northeast Sun Grant Competitive Research Program.

Kebede, E.A. (PI). Key Drivers of Persistent Economic Water Scarcity in Nigeria. CUAHSI Pathfinder Graduate Student Fellowship.

2024 **Kebede, E.A. (PI).** Global Irrigation, Water Scarcity, and Food Sustainability in the 21st Century. AGU Horton Research Grant.

CONFERENCE PRESENTATIONS

2025 **Kebede, E.A.,** Xie, Y., Laboy, G., Bhimani, K., Boser, A., Urfels, A., Khan, B., Adepoju, M., Adeluyi, O., Brauman, K. A., Casirati, S., Chandrasekaran, S., Deines, J. M., Flach, R., Hansen, M.C., Hartman, S., Jobbagy, E., Khan, A., Kurapati, V., Lark, T., Marquez, J., Michael, H., Nakalembe, C., Ng, K.H., Ng, K.W., Nozari, S., Potapov, P., Rosa, L., Siebert, S., Smith, R., Taufer, M., and Davis, K.F. (2025). Global and Local Risks of Rising Water Scarcity to Food Production and Supply. American Geophysical Union Annual Meeting, December 2025, New Orleans, Louisiana, United States. https://agu.confex.com/agu/agu25/meetingapp.cgi/Paper/1937541

Kebede, E.A., Laboy, G., Olaya, P., Taufer, M., Michael, H., Bhimani, K., Davis, K.F. (2025). High-Resolution Global Mapping of Irrigated Areas Using Remote Sensing and Machine Learning. American Geophysical Union Annual Meeting, December 2025, New Orleans, Louisiana, United States. https://agu.confex.com/agu/agu25/meetingapp.cgi/Paper/1953333

2024 **Kebede, E.A.,** Xie, Y., Laboy, G., Bhimani, K., Boser, A., Urfels, A., Khan, B., Adepoju, M., Adeluyi, O., Brauman, K. A., Casirati, S., Chandrasekaran, S., Deines, J. M., Flach, R., Hansen, M.C., Hartman, S., Jobbagy, E., Khan, A., Kurapati, V., Lark, T., Marquez, J., Michael, H., Nakalembe, C., Ng, K.H., Ng, K.W., Nozari, S., Potapov, P., Rosa, L., Siebert, S., Smith, R., Taufer, M., and Davis, K.F. (2024). Global and Local Risks of Rising Water Scarcity to Food Production and Supply. American Geophysical Union Annual Meeting, December 2024, Washington, DC, United States. (Poster). https://agu.confex.com/agu/agu24/meetingapp.cgi/Paper/1582539.

Laboy, G., **Kebede, E. A.,** Olaya, P., Taufer, M., Michael, H., Bhimani, K., Davis, K.F. (2024). High Resolution (30m) Mapping of Irrigated Areas Using Machine Learning. American Geophysical Union Annual Meeting, December 2024, Washington, DC, United States. (Poster). https://agu.confex.com/agu/agu24/meetingapp.cgi/Paper/1642187.

Kebede, E.A., Oluoch, K. O., Siebert, S., Mehta, P., Hartman, S., Jägermeyr, J., Ray, D., Ali, T., Brauman, K.A., Deng, Q., Xie, W., Davis, K.F. (2024). A Global Open-Source Dataset of Monthly Irrigated and Rainfed Cropped Areas (MIRCA-OS) for the 21st Century. American Geophysical Union Annual Meeting, December 2024, Washington, DC, United States. (Poster). https://agu.confex.com/agu/agu24/meetingapp.cgi/Paper/1581599

2023 **Kebede, E.A.,** Oluoch, K. O., Siebert, S., Mehta, P., Hartman, S., Jägermeyr, J., Ray, D., Ali, T., Brauman, K.A., Deng, Q., Xie, W., Davis, K.F. (2023). A global open-source dataset of monthly irrigated and rainfed cropped areas (MIRCA-OS) for the 21st Century, AgMIP9 Global Workshop, June 2023, New York, United States. https://agmip.org/wp-content/uploads/2023/11/AgMIP9-Report.pdf

Kebede, E.A., Oluoch, K. O., Siebert, S., Mehta, P., Hartman, S., Jägermeyr, J., Ray, D., Ali, T., Brauman, K.A., Deng, Q., Xie, W., Davis, K.F. (2023). A global open-source dataset of monthly irrigated and rainfed cropped areas (MIRCA-OS) of the 21st Century. American Geophysical Union Annual Meeting, December 2023, San Francisco, United States. (Poster). https://agu.confex.com/agu/fm23/meetingapp.cgi/Paper/1304785

Kebede, E.A., Hrozencik, A., Marston, L., Van Meter, K., Wei, D., Davis, K. F. (2023). Optimizing cropland conservation payments to minimize nitrogen surplus in the Mississippi River Basin, American Geophysical Union Annual Meeting, December 2023, San Francisco, United States. https://agu.confex.com/agu/fm23/meetingapp.cgi/Paper/1329124

2022 **Kebede, E.A**., Crosato, A., Paron, P., Sloff, K. 1D morphological adaptation of Lower Zambezi River to dam construction, EGU General Assembly 2022, Vienna, Austria, May 2022, EGU22-13022. https://doi.org/10.5194/egusphere-egu22-13022

Kebede E.A., Vasileva S.V., Bojinov, B., Dengiz, O. (2022). Comparing Remote Sensing and In Situ Analyses of Test Wheat Plants as Means for Optimizing Data Collection In Precision Agriculture. II International Cappadocia Scientific Research Congress, Cappadocia-Nevşehir, Turkey.

2021 **Kebede E.A.,** Vasileva S.V., Bojinov, B., Dengiz, O. (2021). Artificial intelligence and remote sensing as a tool for sustainable agriculture: A review. International Soil Science Symposium on Soil Science and Plant Nutrition, Samsun, Turkey.
https://www.fesss.org/upload_pic/926d186a1bf668aad5fb8e525b269ca9.pdf

2017 **Kebede, E.A.,** Kebede, A. (2017). A Comparison of Hydrological Models Under Climate Change on the Water Resources of Megech River Catchment, Abbay Basin, Ethiopia. 4th International Conference on the Advancements of Science and Technology (ICAST - 2017), Bahir Dar, Ethiopia. (Poster).

2016 **Kebede, E.A.,** Kebede, A. (2016). Assessment of Climate Change Impacts on the Water Resources of Megech River Catchment, Abbay Basin, Ethiopia. National Conference on Water Use, Recyclability, and Treatment Technology, Haramaya University, Ethiopia.

FELLOWSHIPS AND AWARDS

2026	MIRCA-OS selected for AGU's "Impactful Datasets in Earth, Space, and Environmental Sciences" collection (AGU Advances, doi:10.1029/2026AV002623)
2025	University of Delaware Open Data Impact Award
2023	CUAHSI Hydroinformatics Innovation Fellowship
2022-2026	Graduate Research Assistantship, Department of Geography and Spatial Sciences, University of Delaware, Newark, Delaware
2020-2022	Erasmus Mundus Joint Master's Degree Scholarship, European Union
2019-2021	Nuffic-Orange Knowledge Program Fellowship, the Kingdom of the Netherlands
2015	Haramaya University Master's Thesis Research Grant, Ethiopian Ministry of Education

TEACHING

University of Delaware (2025-2026)
GEOG 372 — Introduction to GIS (Instructor of Record) Summer 2026
GEOG 271 — Introduction to Geographic Data Analysis (Instructor of Record) Winter 2025

Debre Tabor University, Ethiopia (2017– 2022)

- Soil and Water Conservation (Fall 2017 and 2018)
- Introduction to Natural Resource Management (Spring 2017 and 2018)

Woldia University, Ethiopia (2012– 2017)

- Surface and Ground Water Hydrology (Spring 2015 and 2016)
- GIS and Remote Sensing (Fall 2015 and 2016)
- Research Methods and Experimental Design (Fall 2015 and 2016)
- Irrigation Engineering (Fall 2015 and 2016)
- Farm Machinery (Spring 2012)
- Soil Physics (Spring 2012)
- Introductory Soils (Fall 2012)

Bahir Dar Polytechnic College, Ethiopia (2010– 2012)

- Design of Water Supply System Structures and Construction (Fall 2011 and 2012)
- Design and Operations of Irrigation Structures (Spring 2011 and 2012)

MENTORING AND STUDENT SUPERVISION

2023 Gabriel Laboy. Mentor. Mapping of Irrigated Areas Using Machine Learning. Summer Intern at Davis Lab. [Currently a graduate student at Arizona State University, Arizona]

2022 Endalamaw Dessie. Academic advisor and mentor. Erasmus Mundus Joint Master's Degree in Soil Science (emiss) Student at Ondokuz Mayis University.

2020 Ashenafi Tadesse Marye. Mentor. [A PhD student at Syracuse University, New York]

2019 Yonas Gebresilase. Undergraduate thesis advisor. The impact of Land Cover change on the hydrology of the Ribb Watershed. [Currently an MSc student at the University of Dayton, Ohio]

2016 Wudu Abiye. Undergraduate thesis advisor and mentor. Review of the Impacts of Land Degradation on Agricultural Production in Ethiopia. [Currently a Marie-Curie doctoral fellow in Europe]

2012 Zelalam Regassa Womber. Academic advisor and mentor. [PhD student at the University of Massachusetts Amherst]

ACADEMIC LEADERSHIP AND SERVICE

2017-2019	Member of research grant review committee, Guna Tana Integrated Field Research and Development Center, Debre Tabor University, Debre Tabor, Ethiopia
2018-2019	Member of the national conference organizing committee, Debre Tabor University, Debre Tabor, Ethiopia
2015-2017	Member of the Academic Council (AC), College of Agriculture, Woldia University, Woldia, Ethiopia
2012-2013	Interim head of Soil and Water Resource Management Department, Woldia University, Ethiopia
Ongoing	Reviewer: Nature Water, Nature Communications, Scientific Data, and other peer-reviewed outlets

SKILLS

GIS and Remote Sensing
ArcGIS | QGIS | Google Earth Engine | Erdas Imagine| ENVI| SNAP

Data analysis tools
Python | R| Machine Learning| Deep Learning| Cloud Computing| HPC

Hydrological and Hydraulic Modeling
HEC-RAS | HEC-HMS| SWAT | GR4J | HBV-Light | MIKE-SHE | MIKE-11

Crop Water Productivity and Other Agricultural Models
CROPWAT | AquaCrop |RETC-Soil Water Potential Model |FAO-WAPOR Model| WEAP

GitHub account: https://github.com/EndiKebede

AFFILIATIONS

American Geophysical Union
European Geosciences Union
International Water Resources Association
American Water Resource Association
National Ground Water Association (US)
Ethiopian Geophysical Union

SELECTED MEDIA COVERAGE

2025 Research featured in UDaily, University of Delaware Global Crop Data

UD Honors Excellence in Open Data

2024 Research featured in UDaily, University of Delaware. Global Food Production Data Scarcity

Research featured on Phys.org. The study looks at challenges facing global food production data

Research featured on the Cornell College of Agriculture and Life Sciences. New Study Highlights Challenges in Global Food Production Data

2023 Featured in UDaily, University of Delaware. Solving Water Scarcity